

Simulation Problem (80 points):

Convert the design in HW4 to a differential amplifier with single-ended output.

Consider load resistor $R_D=500\Omega$, the capacitor C_{load} of 2pF, and $V_{dd}=1.3V$.

You are to use an 130nm CMOS process. The design goals for this circuit are as follows:

Low frequency differential-in to single-ended output ac voltage gain > 30dB

(ie. $\text{Gain} = V_{out}/(V_{in+}-V_{in-})$)

Maximum output swing > 0.3Vpk-pk-se

Unity gain bandwidth > 500MHz

What you need to turn in for this problem are:

- (20 pts) Hand calculation results including the values of DC bias voltage and current, transistor size, small-signal gain, output resistance, and maximum output swing.
- (20 pts) Cadence simulation results including a plot of small-signal gain and phase, and information showing DC conditions of the MOSFET.

Report: Attach schematics with DC operating points, AC gain and phase, transient simulation showing results at the bias points with the output and input swings. You have to ensure that none of these transistors go out of saturation at the maximum or the minimum input voltages. V_{bias} , V_{b2} , and V_{b3} are implemented using VDC symbol in analogLib library in cadence.

